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Updating: Perspectives for HET physics studies at a muon-hadron collider at CERN

Hamzeh Khanpour

(on behalf of HELP4CERN Study Group)

Faculty of Physics and Applied Computer Science
AGH University of Krakow

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Outline

1. Physics motivation and benchmarks
2. HET overview: Higgs, electroweak and top
3. High-energy $\gamma\gamma$ physics at LHμC
4. Top physics at LHμC
5. Beyond SM and long-term outlook

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Why a muon–hadron collider for HET physics?

- A muon-hadron collider combines the clean lepton-hadron initial state of DIS with a multi-TeV centre-of-mass energy, opening a new regime for Higgs, electroweak, top, and BSM studies.
- Relative to an electron-proton collider, replacing the electron beam by a muon beam strongly relaxes the synchrotron-radiation limitation, allowing significantly higher beam energies and hence much larger hard-scattering rates, and pushes the centre-of-mass energy deep into the multi-TeV regime.
- Relative to proton-proton collisions, the μp environment remains experimentally cleaner, with smaller QCD backgrounds for many precision and BSM channels.

Three complementary physics paths at LHμC

LHμC combines three powerful and complementary directions: a strong QCD programme, high-precision Higgs/top/electroweak measurements, and a powerful probe of physics beyond the Standard Model.

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Benchmark working points used in this talk

- The present talk focuses on the physics case of the LHμC, using the current HELP4CERN working benchmarks:
- The nominal baseline is

$$E_\mu = 500\ \mathrm{GeV},\ E_p = 7\ \mathrm{TeV},\ \sqrt{s} \simeq 3.7\ \mathrm{TeV},$$

with a $\mathcal{L}_{\mathrm{int}} = 250\ \mathrm{fb}^{-1}$.

- The main upgrade scenario is

$$E_\mu = 1\ \mathrm{TeV},\ E_p = 7\ \mathrm{TeV},\ \sqrt{s} \simeq 5.3\ \mathrm{TeV},$$

with a $\mathcal{L}_{\mathrm{int}} = 1000\ \mathrm{fb}^{-1}$.

LHμC physics message

These working points are sufficient to demonstrate the main message of LHμC: already at the 500 GeV baseline, LHμC enters the multi-TeV regime and becomes highly competitive for Higgs, top, electroweak, $\gamma\gamma$, and representative BSM studies.

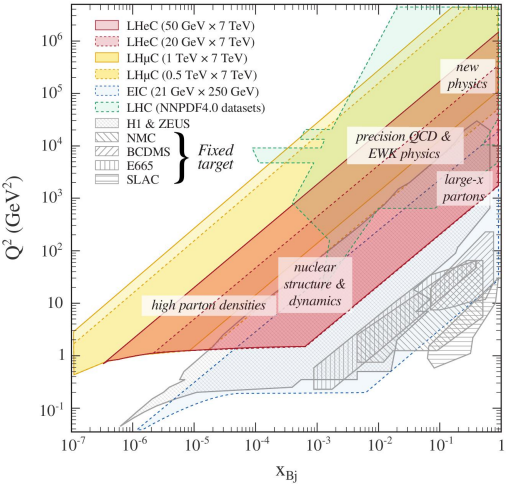
Benchmarks

Machine	**Benchmark**
LHeC	$E_e = 20,\ 60$ GeV
LHμC baseline	$E_\mu = 500$ GeV
LHμC upgrade	$E_\mu = 1$ TeV
Configuration	**Value**
$\sqrt{s}_{\mu p}$ (baseline)	$\simeq 3.7$ TeV
$\sqrt{s}_{\mu p}$ (upgrade)	$\simeq 5.3$ TeV
$\mathcal{L}_{\mathrm{int}}$ (baseline)	$250\ \mathrm{fb}^{-1}$
$\mathcal{L}_{\mathrm{int}}$ (upgrade)	$1000\ \mathrm{fb}^{-1}$

HE*l*-*p*4CERN

See talk by Krzysztof Piotrzkowski

Kinematic reach and QCD physics landscape



Available (x, Q^2) plane for the proposed LHμC, compared with fixed-target, collider, and EIC kinematic coverage.

QCD program at LHμC

- The multi-TeV μp configuration extends the DIS kinematic coverage well beyond HERA and significantly beyond the LHeC.
- The same machine that reaches extreme x and Q^2 also strengthens the QCD programme through improved access to proton PDFs, and precise determinations of α_s.
- This motivates the broader CERN bridge-project narrative: LHμC strengthens both the QCD frontier and the HET/BSM programme.

See talk by Piotr Kotko

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Representative HET benchmark cross sections

What does this table show?

LHμC as a genuine HET factory: Higgs production is large enough to support precision Higgs physics, including benchmark coupling studies & the top sector benefits the most, enabling both precision top measurements and sensitivity to rare top processes, & even the 500 GeV baseline opens a qualitatively new regime for electroweak, EFT, and BSM physics.

Benchmark inclusive cross sections

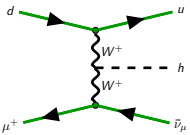
Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(\ell p \to \nu_\ell H + \text{jet} + X)$ [fb]	52	164	438	731
$\sigma(\ell p \to \ell H + \text{jet} + X)$ [fb]	5.6	19	135	221
$\sigma(\ell p \to \ell\, t\bar{t} + X)$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(\ell p \to \nu_\ell t + X)$ [pb]	0.55	2.89	21.26	36.77

Key comparisons

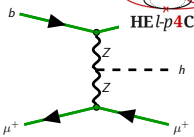
Relative to the 50 GeV LHeC benchmark, the 500 GeV LHμC increases rates by approximately: ~ 2.7 for charged-current Higgs production, ~ 7 for single-top production, and ~ 32 for $t\bar{t}$ production.

Higgs production at LHμC: channels, rates, and statistics

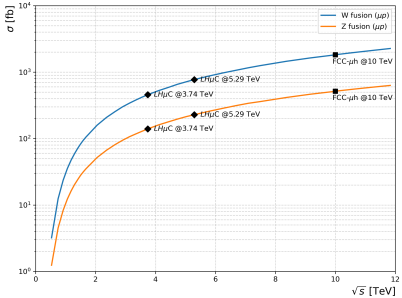
- Higgs production is dominated by CC VBF, with a subleading but still sizeable NC contribution.
- Using the current benchmarks:
  - 500 GeV LHμC: $\sigma_{\rm CC} \simeq 438$ fb, $\sigma_{\rm NC} \simeq 135$ fb,
  - 1 TeV LHμC: $\sigma_{\rm CC} \simeq 731$ fb, $\sigma_{\rm NC} \simeq 221$ fb.
- For the benchmark integrated luminosities, these correspond to roughly:
  - 1.10×10^5 CC Higgs and 3.38×10^4 NC Higgs events at 500 GeV,
  - 7.31×10^5 CC Higgs and 2.21×10^5 NC Higgs events at 1 TeV.
- This already motivates precision Higgs-coupling studies beyond a mere rescaling of LHeC expectations.



CC: $\mu^+ p \to \bar{\nu}_\mu\, j\, h$



NC: $\mu^+ p \to \mu^+\, j\, h$



Higgs boson production cross sections at LHμC for the CC (W-fusion) and subleading

Precision Higgs case study: $h \to b\bar{b}$ and coupling prospects

- A concrete benchmark already exists for the charged-current VBF process with $h \to b\bar{b}$.
- The parton showering, fast detector simulation, and a cut-based analysis were used to estimate:
  - $\Delta\sigma/\sigma \simeq 1.19\%$ and 0.69% for the two μp collider setups,
  - corresponding benchmark sensitivities of about 0.77% and 0.60% on g_{hbb}.
- This strongly supports the claim that LHμC can be a competitive Higgs factory.

	μp-1	μp-2
$\sigma_{\rm sig}$ [pb]	0.24	0.39
$\sigma_{\rm bgd\text{-}Z}$ [pb]	0.30	0.50
$\sigma_{\rm bgd\text{-}mj}$ [pb]	357.9	563.8
$\epsilon_{\rm cut}^{\rm sig}$	0.22	0.21
$\epsilon_{\rm cut}^{\rm bgd\text{-}Z}$	0.025	0.025
$\epsilon_{\rm cut}^{\rm bgd\text{-}mj}$	9.7×10^{-5}	1.23×10^{-4}
$\Delta\sigma/\sigma$	1.19%	0.69%
$\Delta g_{hbb}/g_{hbb}$	0.77%	0.60%

Table: Benchmark $h \to b\bar{b}$ case study at μp colliders: signal and background cross sections, cut efficiencies, and projected sensitivities for the Higgs coupling to bottom quarks.

Benchmark precision Higgs case: $h \to b\bar{b}$ in CC VBF

- A concrete benchmark study already exists for the charged-current VBF channel

$$p\mu^+ \to j\,\bar{\nu}_\mu\,h, \qquad h \to b\bar{b}.$$

- Analysis chain used in this study:
  - MadGraph5_aMC@NLO + Pythia6 + FastJet + Delphes,
  - LHeC-like asymmetric detector setup and 75% b-tagging efficiency,
  - selection with exactly two b-jets and

$$|m_{bb} - m_h| < 25\ \mathrm{GeV}.$$

- This gives a direct benchmark on the measurable rate

$$\sigma(p\mu^+ \to j\,\bar{\nu}_\mu\,b\bar{b}) \;\propto\; \frac{g^2_{hWW}\,g^2_{hbb}}{\Gamma_h},$$

	LHμC-500	LHμC-10(
σ_{sig} [pb]	0.24	0.39
$\sigma_{\mathrm{bgd\text{-}Z}}$ [pb]	0.30	0.50
$\sigma_{\mathrm{bgd\text{-}mj}}$ [pb]	357.9	563.8
$\Delta\sigma/\sigma$	1.19%	0.69%
$\Delta g_{hbb}/g_{hbb}$	0.77%	0.60%

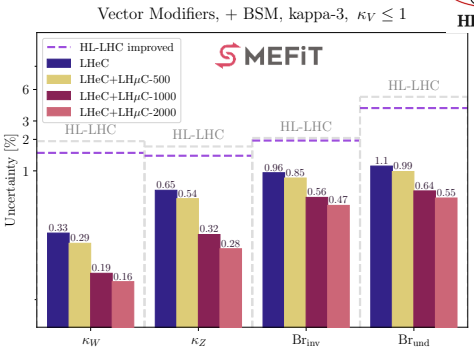
Benchmark charged-current $h \to b\bar{b}$ sensitivities in the current HELP4CERN study. The quoted coupling reach follows from the benchmark rate measurement after propagating the external uncertainty on g^2_{hWW}/Γ_h.

Broader Higgs-coupling reach: exploratory global-fit picture

- Beyond the dedicated $h \to b\bar{b}$ benchmark, the current white-paper draft also includes an exploratory Higgs coupling fit in the **kappa-3** framework.
- The compared scenarios are:
  - HL-LHC,
  - HL-LHC improved,
  - LHeC,
  - LHeC + LHμC-500 / 1000 / 2000.
- The strongest visible improvement is in the **vector sector**:

$$\delta\kappa_W \simeq 0.19\% \ (\mathrm{LH}\mu\mathrm{C}\text{-}1000), \qquad 0.16\% \ (\mathrm{LH}\mu\mathrm{C}\text{-}2000).$$

- The companion fermion and loop panels show further, gener-



Exploratory vector-modifier result in the kappa-3 framework. The panel compares HL-LHC-based benchmarks with LHeC and with LHeC+LHμC stages. The most important message for the talk is the strong projected improvement in the vector sector, especially for κ_W.

Electroweak programme and EFT opportunities

- Beyond Higgs rates, LHμC offers a broader electroweak programme based on high-Q^2 DIS, vector-boson fusion, and photon-induced reactions.
- The larger kinematic span relative to the LHeC is expected to strengthen sensitivity to precision electroweak parameters such as m_W and $\sin^2\theta_W$, while also extending the SMEFT lever arm.
- To explicitly highlights:
  - precision electroweak measurements,
  - orthogonal SMEFT directions relative to pp and e^+e^- machines,
  - tests of lepton universality,
  - and photon-induced EFT sensitivity in $\gamma\gamma \to W^+W^-$, ZZ, and heavy final states.

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High-energy $\gamma\gamma$ physics is a natural part of the LHμC programme

- LHμC naturally operates as a high-energy $\gamma\gamma$ collider through quasi-real photons radiated from the incoming muon and proton beams.
- In the 500 GeV baseline, the effective $\gamma\gamma$ centre-of-mass energy extends into the multi-TeV regime, opening a broad programme of clean exclusive measurements.
- This part of the physics case is directly relevant to the HET talk because it connects to:
  - anomalous τ moments,
  - $\gamma\gamma \to W^+W^-$ and quartic gauge couplings,
  - loop-induced $\gamma\gamma \to ZZ$,
  - exclusive $t\bar{t}$ production,
  - and BSM charged-particle pair production.

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Flagship $\gamma\gamma$ channel I: $\tau^+\tau^-$ and anomalous τ moments

- The elastic $\gamma\gamma \to \tau^+\tau^-$ rate at LHμC is already very large:
  - $\sigma_{\rm el} \simeq 58.56$ pb at 3.7 TeV,
  - rising to about 70.3 pb for the 5.3 TeV upgrade benchmark.
- This delivers a high-statistics laboratory for precision studies of the $\gamma\tau\tau$ vertex and for anomalous magnetic / electric dipole moment analyses.
- The central phenomenological idea is simple: anomalous contributions harden the high-$M_{\tau\tau}$ tail, so the invariant-mass spectrum becomes the main discriminating observable.
- This is one of the cleanest examples showing how LHμC combines high energy with a low-background, exclusive topology.

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Flagship $\gamma\gamma$ channel II: W^+W^- and anomalous quartic gauge couplings

- The elastic $\gamma\gamma \to W^+W^-$ cross section grows from about 0.037 pb at the LHeC to about 0.166 pb at LHμC@3.7 TeV and 0.258 pb at the 5.3 TeV upgrade benchmark.
- The EFT signal is not mainly an overall rate effect: the real sensitivity comes from the **high-M_{WW} tail**, where dimension-8 operators enhance the spectrum.
- Relative to the LHC, the μp environment offers cleaner event topology, more precise kinematics, and potentially simpler control of proton/muon tagging.
- This makes LHμC a serious laboratory for aQGC studies rather than merely an auxiliary by-product of the machine.

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Flagship $\gamma\gamma$ channels III: ZZ, $t\bar{t}$, and heavy exclusive final states

- Loop-induced $\gamma\gamma \to ZZ$ is especially interesting because it is absent at tree level in the Standard Model and therefore provides a clean probe of high-mass electroweak dynamics.
- Representative elastic cross sections at 3.7 TeV are:
  - $\sigma_{\rm el}(\gamma\gamma \to ZZ) \simeq 0.284$ fb,
  - $\sigma_{\rm el}(\gamma\gamma \to t\bar{t}) \simeq 0.486$ fb.
- Although smaller than the $\tau\tau$ and WW cases, these channels gain importance because they probe loop structure, heavy thresholds, and the onset of high-energy electroweak / top dynamics.
- The same photon-induced framework also extends to exclusive pair production of BSM charged states such as higgsinos and sleptons.

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LHμC as a top factory

- The most striking HET advantage of LHμC relative to the LHeC is the **top sector**.
- Inclusive benchmark rates:
  - single top: 21.26 pb at 500 GeV and 36.77 pb at 1 TeV,
  - $t\bar{t}$: 1.03 pb at 500 GeV and 2.23 pb at 1 TeV.
- For the benchmark integrated luminosities, this translates into:
  - $\sim 5.3 \times 10^6$ single-top events at 500 GeV,
  - $\sim 2.6 \times 10^5$ $t\bar{t}$ events at 500 GeV,
  - $\sim 3.7 \times 10^7$ single-top events at 1 TeV,
  - $\sim 2.2 \times 10^6$ $t\bar{t}$ events at 1 TeV.
- This is why the current collaboration narrative repeatedly describes LHμC as a genuine top factory.

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Top event topology, acceptance, and why LHμC helps experimentally

- Beyond inclusive rates, dedicated semileptonic $t\bar{t}$ studies indicate that the LHμC event topology is more central than the LHeC one.
- This matters because the gain at generator level is not purely from larger cross sections; it is also supported by strong kinematic acceptance.
- Representative semileptonic acceptance numbers show only modest degradation even for tighter lepton rapidity cuts, with the LHμC case remaining particularly favourable.
- Therefore the larger top rates are not merely formal parton-level advantages: they are likely to remain exploitable after realistic selections.

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Rare top, FCNC, and EFT directions

- The top sector at LHμC is interesting not only for statistics, but also for **rare processes and anomalous couplings**.
- Dedicated studies already show that top-FCNC sensitivities improve substantially from the LHeC to the LHμC, reflecting both the higher rates and the harder kinematics.
- The characteristic μp topology provides a useful environment for studying anomalous $tq\gamma$, tqZ, and tqH interactions, as well as broader EFT deformations in single-top and associated channels.
- In the presentation, this part should be framed as one representative example of how the enormous top samples can be converted into BSM reach.

Why LHμC is also a distinctive BSM machine

- Muon–proton collisions combine a point-like lepton beam, high centre-of-mass energy, and the possibility of **resonant single production** of new states that are more difficult to isolate at hadron colliders.
- The LHμC is naturally powerful for BSM scenarios tied to second-generation leptons.
- Representative examples include:
  - R-parity-violating squark searches,
  - color-octet muons (leptogluons),
  - excited leptons,
  - contact interactions,
  - and other muon-coupled new-physics scenarios.
- The talk should keep this part selective: one or two benchmark examples are enough to make the point.

Kingman Cheung and Zeren Simon Wang, *Physics potential of a muon-proton collider*, Phys. Rev. D 103 (2021) 116009.

Representative BSM benchmarks: RPV SUSY and color-octet muons

- **RPV SUSY:** The μp collisions can probe resonantly produced squarks with coupling reaches stronger than HL-LHC bounds in the on-shell regime.
- **Color-octet muons:** A 5σ discovery reach of about:
  - ~ 2.3 TeV at the HL-LHC,
  - ~ 3.0 TeV for the 500 GeV LHμC option,
  - ~ 4.2 TeV for the 1 TeV LHμC option.
- These examples are sufficient to demonstrate that the BSM case is not generic hype: it is tied to concrete channels where the μp initial state is a genuine advantage.

Kingman Cheung and Zeren Simon Wang, *Physics potential of a muon-proton collider*, Phys. Rev. D 103 (2021) 116009.

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FCC-hh era outlook and final take-home points

- The long-term perspective goes well beyond the 3.7 TeV baseline.
- The FCC-hh era is explicitly connected to a future high-energy μp option, turning the current LHμC concept into a stepping stone toward even more powerful HET and BSM programmes.
- In particular, a future FCC-μh machine would extend:
  - Higgs statistics and coupling reach,
  - top and rare-top production,
  - high-mass electroweak and $\gamma\gamma$ studies,
  - and resonant BSM sensitivity.

HET@LHμC

(1) LHμC is a serious HET factory;
(2) the **top sector** is one of its clearest near-term strengths;
(3) Higgs, EW, EFT, and $\gamma\gamma$ studies are all strongly motivated;
(4) the concept also provides a natural bridge toward FCC-era μp physics.

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